Generating a Low-code Complete Workflow via Task Decomposition and RAG

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***Abstract*—AI technologies are moving rapidly from research to production. With the popularity of Foundation Models (FMs) that generate text, images, and video, AI-based systems are increasing their complexity. Compared to traditional AI-based software, systems employing FMs, or GenAI-based systems, are more difficult to design due to their scale and versatility. This makes it necessary to document *best practices*, known as design patterns in software engineering, that can be used across GenAI applications.**

Our first contribution is to formalize two techniques, Task Decomposition and Retrieval-Augmented Generation (RAG), as design patterns for GenAI-based systems. We discuss their trade-offs in terms of software quality attributes and comment on alternative approaches. We recommend to AI practitioners to consider these techniques not only from a scientific perspective but also from the standpoint of desired engineering properties such as flexibility, maintainability, safety, and security.

As a second contribution, we describe our industry experience applying Task Decomposition and RAG to build a complex real-world GenAI application for enterprise users: *Workflow Generation*. The task of generating workflows entails generating a specific plan using data from the system environment, taking as input a user requirement. As these two patterns affect the entire AI development cycle, we explain how they impacted the dataset creation, model training, model evaluation, and deployment phases.

***Index Terms*—generative AI, RAG, task decomposition, workflows.**

I. Introduction

Since the introduction of ChatGPT two years ago, more and more systems based on Generative AI (GenAI) are being shipped to consumers. The underlying technology is a Foundation Model (FM), more commonly a Large Language Model (LLM), able to generate text, images, and, more recently, video.

Due to current limitations in FMs, such as their black-box nature and sensitivity to data distribution shifts [1], [2], building systems leveraging their capabilities is particularly challenging. Integrating AI models into software systems already adds a new set of complex design choices to software engineering [3]–[7]. This complexity has increased as the range of possible GenAI-based systems has grown significantly due to their increasing abilities to generate almost any kind of output. For instance, whereas in the past AI models added classification or regression capabilities to software, nowadays systems include more sophisticated features such as code completion, explanation, and search [8].

The software engineering community has recognized the need for new design patterns for AI-based systems to provide blueprint solutions to the challenges involved in integrating AI models [4], [9], [10]. In particular, patterns for architectural design decisions are well-documented [5], [11]–[13].

Generative AI requires adapting existing design patterns for AI-based systems and adding new ones, as problems specific to generative models such as hallucination (safety) or large number of output tokens (scalability) can affect software systems in new, unexpected ways [14]. It is also not trivial to engineer these systems to allow adding features rapidly (modifiability) and to allow them to interact effectively with other non-AI components (interoperability).

As AI practitioners are responsible for building the GenAI models (fine-tuning or prompting), they need to be aware of the impact, positive and negative, that modern techniques have on software engineering [15]. However, the typical process to build AI-based systems still sees a rift between scientists and engineers, where the former develop the models and the latter bring the models into production [5], [16].

The consequence is that models are built with little regard to the desired quality attributes of the overall system, as the main concern of model building is to maximize metrics to prove that the solution works. This may increase the time to deploy the AI-based system into production due to possible redesign of datasets and model rebuilding if the pipeline proves to be too slow, unsafe, etc. Therefore, it is necessary to discuss how common AI techniques affect software engineering.

To help bridge this gap between AI and software engineering, the first part of this paper formalizes two common techniques as design patterns since their utilization impacts the entire engineering effort, crucially the architectural design of the system. Their impact goes beyond constructing the model; hence it is important to discuss how they affect the flexibility, maintainability, safety, and security of the system, among other commonly desired software quality attributes [17].

Task Decomposition is an application of *divide and conquer* to the AI domain. This technique allows breaking a machine learning (ML) task into sub-tasks, which can be more easily solved. It has been proposed for a variety of tasks such as speech processing [18], film trailer generation [19], and abstract visual reasoning [20].

Retrieval-Augmented Generation (RAG) is a well-known technique to allow an FM to interact with data from its

environment so that its knowledge is no longer limited by what is stored in its model weights [21]. RAG has been used with LLMs to reduce hallucination [22], to give them access to world knowledge [23], and to make the generated content more traceable to its sources [24].

As our formalization of Task Decomposition and RAG as design patterns derives from our experience in industry, the second part of this paper offers a case study of how we built a GenAI application called *Workflow Generation*. The ML task of generating workflows is similar to generating structured text such as code. A workflow can be seen as a *very specific plan* that automates tasks by interacting with its environment. Since the resulting application was deployed to end-users as part of an enterprise system, it had to meet particular engineering constraints and achieve certain quality attributes.

Using a case study is appropriate as how design patterns are applied depends heavily on the context and constraints in which the system is built. The objective is to show that Task Decomposition and RAG can result in well-engineered systems. We followed traditional guidelines for case studies in software engineering [25] to report how these two techniques resulted in good design decisions along the AI development cycle, from data labeling to model deployment. As we are conscious that one can only make limited conclusions from one case study, we invite other AI practitioners to share their work when using these techniques.

Our contributions are the following:

- We formalize Task Decomposition and RAG as design patterns, documenting and discussing how they impact the engineering of GenAI-based systems.
- We offer a case study of how we applied these two techniques to build *Workflow Generation*, a GenAI application, to illustrate their positive impact along the entire development cycle.

II. Related Work

There is a rich body of literature on **software engineering for ML**, ranging from case studies [6], [7] to surveys drawn from academic and gray literature [5], [10], [16]. Some papers contribute a list of design patterns for AI-based systems [4], [9]. A limitation of literature reviews is that they necessarily lack detail, and thus their description of design patterns are at a high-level, divorced from the context in which they are applied. Moreover, most of the existing list of design patterns for ML dates from the pre-generative AI era, when the AI components were relatively simple compared to current uses of FMs. Our work complements this body of work by discussing two techniques in the context of GenAI and providing a detailed description of how we applied them in industry.

Foundation Models are being heavily studied by AI researchers and there is recent work aiming to document how they are currently being inserted in the software stack [26]. As FMs are the fundamental AI building block of modern GenAI-based systems, there are several architectural design choices to make, such as whether to use an FM hosted internally or hosted by an external provider such as OpenAI [27]. There are nascent efforts underway to document the new set of challenges FMs add to software engineering [14]. Our paper fits into this line of work by discussing how FMs can be used in a real-world application with Task Decomposition and RAG.

Task Decomposition as a technique to solve complex ML tasks is not new. It is related to AI design patterns such as Two-Phase Predictions, AI pipelines, and Multi-Layer Pattern [4]. In the GenAI literature, it is related to the Coordination connector, where the FM coordinates tasks between AI and non-AI components [26], and it can be implemented with a Chain of FMs where each FM is responsible for executing a sub-task that is part of a larger task [27]. We expand on this work by discussing the software quality attributes that this technique offers to GenAI-based systems.

Most modern GenAI systems employ an implementation of **Retrieval-Augmented Generation**, as a way to improve the output quality of FMs. RAG can be used in an iterative, recursive, or adaptive fashion [21], and to adapt FMs to specialized domains [28]. It can be seen as an implementation of the Strategy pattern in the context of AI [4], where the strategy is made up of two interfaces: one to generate text and one to retrieve suggestions used in the generation process. The generation is affected by the quality of the retrieved results, whereas the retrieval is affected by the quality of the query, which can be user-provided or FM-generated. As with Task Decomposition, we formalize its usage as a design pattern and discuss its impact on software quality attributes.

Lastly, automatically **generating workflows** is an important task for enterprise systems, as building them generally requires expert domain knowledge. Workflows provide significant value to end-users because they automate repeatable processes in several domains such as IT (e.g., acquiring an IT asset) or HR (e.g., onboarding new employees). There exist several approaches to generate them, such as having an LLM generate an initial plan, then asking users to correct the generated plan, followed by asking the LLM again to regenerate the workflow and its details [29]. In our use case, we generate a very detailed plan with environment artifacts without any additional input from the user besides the initial requirement. The workflow is generated in a code-like representation that can be rendered in a web user interface.

III. GenAI Design Patterns

We consider Task Decomposition and RAG as design patterns because they are reusable solutions to the problem of integrating GenAI components into larger software systems. While they affect all phases of the AI development cycle, they crucially affect the architecture of the system.

Inspired by the template used to document design patterns in the seminal work by Gamma et al. [30], we describe these two techniques following most of the template's sections, and motivate them using classical software quality attributes [17].

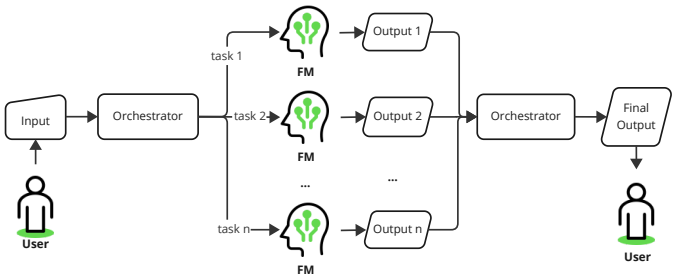
A. Task Decomposition

- *Intent*: Break a complex ML task into simpler sub-tasks.

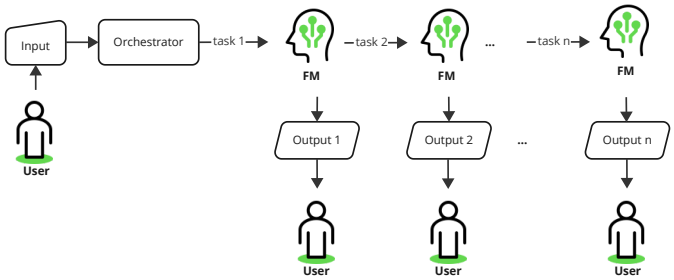
- *Also Known As*: Two-Phase Predictions, AI Pipelines, Multi-Layer Pattern, Chain of FMs.
- *Motivation*:
  1) Scalability: FMs can output large number of tokens per input, creating performance risks as the output size increases. Asking the FM to generate output for smaller sub-tasks helps limit scalability issues.
  2) Modularity: By reducing the granularity of the problem, different FMs can solve different parts of the problem, improving the system maintainability.
  3) Functional correctness: As with humans, breaking a problem into sub-problems can aid the FM increase the quality of its output.
  4) Time behaviour: Generating output at the sub-task level reduces the time it takes for end-users to receive a response.
  5) Modifiability: As the model can handle more functionality due to more labeled data or better FMs, more sub-tasks can be added without breaking the overall system design.
  6) Testability: Evaluating simpler sub-tasks instead of one complex task makes testing FMs easier.
  7) Analysability: It is easier to diagnose problems in the generated output if generation takes place incrementally via sub-tasks.
- *Applicability*: Task Decomposition is suitable for ML tasks that require complex structured output or output that can be structured, such as code generation or essay writing. In the former, when generating a class, the AI model can first generate the signatures of the functions of the class, and then generate the content of each function. In the latter, the AI model can first generate the leading sentence of each paragraph, and then generate each paragraph. In these cases, asking the model to generate the entire output at once may result in lower correctness and/or delays in responses.
- *Structure*: Figure 1 shows sample structures.
- *Participants*: One or more FMs which execute the sub-tasks, and another component (AI or non-AI) to orchestrate how the sub-tasks are distributed.
- *Collaborations*: The orchestrator component divides the sub-tasks among FMs so that the main task is completed. This component can be another FM (e.g., AI agents) or traditional software code (i.e., non-AI component) that calls the FMs deterministically. There also needs to be a mechanism to create the sub-task input, which can come from FMs or non-AI components.
- *Consequences*:
  1) Decomposition can lead to lower response times and improved system scalability as the size of the FM output is decreased significantly. However, the overall system response time may increase due to the task orchestration.
  2) By having specialized FMs or specialized prompts for the same FM, the overall correctness and modularity of the solution is increased. If there is a sub-task that is more difficult or requires more training/prompting/tuning, then more time and resources can be devoted to it. However, the system complexity increases as there are more components to maintain. For instance, instead of one single prompt, there are several to be maintained and updated. This additional work can be justified only when the ML task is sufficiently complex.
  3) Instead of having to test the overall ML task, test cases or evaluation sets can be created according to the sub-tasks. Error analysis and remediation steps can be conducted according to the different sub-tasks. On the other hand, integration tests are still needed as the output of one sub-task, even if better at the sub-task level, can impact the quality of another sub-task in unforeseen ways.
  4) By making sub-tasks the unit of shippable sub-features, then the system can be adapted to support more and more functionality. However, knowing how to align sub-tasks to sub-features requires deep knowledge of the use case.
- *Implementation*: Decomposing the task may affect data labeling both for training and evaluation. If a dataset is provided at the task level, then it would have to be repurposed so that it can be used for the sub-tasks. Additionally, model deployment can become more complex if more than one FM is used.
- *Related Patterns*: The opposite of using Task Decomposition is to let the FM generate the output for the task all at once. This is suitable for ML tasks that are simple or difficult to break into sub-tasks. A pattern related to Task Decomposition is AI Agents, where an FM decomposes the task into sub-tasks and orchestrates their completion (orchestrator agent), and other FMs solve the sub-tasks (worker agents) [31].

B. Retrieval-Augmented Generation

- *Intent*: Provide environment data to an FM so that it can adjust its output according to real and available input.
- *Also Known As*: RAG.
- *Motivation*:
  1) Security: FMs can introduce security risks if allowed to call functions or tools available in the environment [32], [33]. By suggesting environment data in the input to the FMs, they can interact with the environment in a more secure way.
  2) Safety: Retrieving and suggesting facts to an FM during generation can decrease the amount of hallucination in their output.
  3) Modularity: To simplify the ML task for the FM, it is desirable to separate *how* the task is solved from *what* data is needed to solve it. Requiring the FM to memorize all the data it needs for the task in its model weights is unfeasible.



(a) Task is decomposed into sub-tasks that are executed in parallel. Only the final output is shown to the user.



(b) Task is decomposed into sequential sub-tasks whose output are shown to the user, so that users can see the result incrementally.

Fig. 1: Sample structures of Task Decomposition.

4) Interoperability: Modern AI systems require FMs to interact with artifacts or knowledge available in the environment in which they run.
5) Functional correctness: The knowledge embedded in the model weights of an FM is limited to the data it has received during training. By augmenting its knowledge with information from the environment, it can increase its correctness.
6) Self-descriptiveness: GenAI-based systems tend to be black boxes; showing the retrieval results gives users more visibility on the system's inner workings.
7) Testability: RAG allows evaluating the retriever and the indexed data sources independently of the FM.
8) Analysability: When errors arise, RAG allows a diagnostic of whether the FM made an error due to lack of AI capabilities, or due to inappropriate environment data available to it.

- *Applicability*: RAG is useful when an FM needs to access up-to-date knowledge from its environment, allowing it to generalize to scenarios or input data not seen during its training. When FMs are not grounded to the environment in which they run, they have a higher likelihood of producing answers that are plausible yet false. When models tend to hallucinate, it is a good idea to use RAG.
- *Structure*: Figure 2 shows sample structures.
- *Participants*:
  1) A retriever, normally another AI model, used to encode the data to be retrieved as well as the search query. A simpler keyword search mechanism can be used to reduce the complexity of the solution [34].
  2) An FM which will receive data from the retriever in its input prompt. This FM can also be used to create the queries that will be used to retrieve data from the environment.
  3) Sources of data where the retriever pulls data from: databases, text documents, the web, etc.
- *Collaborations*: There needs to be a mechanism to index the environment data so that it can be efficiently retrieved. The retriever is used to create embeddings, or vectors, that can be indexed offline. At generation time, when retrieval needs to happen, this same retriever will create an embedding for the query. A similarity (e.g. cosine similarity) score will be created to retrieve the indexed data that is most relevant to the query. This data is then passed to the FM during generation. There are several ways to orchestrate this collaboration between retrievers and generators: iterative, recursive, and adaptive [21].
- *Consequences*:
  1) As the FM will receive suggestions of what data it should include in its output, the likelihood of hallucinated and unsafe output decreases. However, this requires the sources of data to be kept up-to-date and free of unwanted output, or having another mechanism to filter out the suggestions.
  2) The responsibilities for providing up-to-date environment data to the FM will rest on the retriever and the indexed sources of data, allowing the FM to be optimized for solving the ML task. This separation of concerns increases modularity and makes it clear that only the retriever AI component interacts with sensitive non-AI components, such as databases. The trade-off is higher system complexity as now there are two AI components to maintain: the retriever and the FM.
  3) The system is likely to be more correct, especially if its functionality depends on access to latest data or knowledge. However, there is no guarantee that the retriever suggestions are of good quality or that the FM will accept them during generation [35]. Possible approaches to ameliorate this include offering many suggestions to the FM instead of one, training the FM to receive suggestions in a particular form, and jointly training the retriever and the FM.
  4) Showing the retrieval suggestions to end-users along with the generated output allows them to decide whether the output is correct. This provides a measure of self-descriptiveness as the generation process is more transparent, especially in knowledge-based ML tasks such as summarization and question-answering. However, showing why the FM used or ignored the suggestions is currently not feasible, as explaining the output of an FM is still an open research question.
  5) The separation of concerns between retrieval and

generation allows testing each of these AI components separately. Retrieval is generally evaluated with metrics such as Recall, Hit Rate or Mean Reciprocal Rank [21] whereas the metrics used to test the overall system output resulting from generation depend largely on the use case.

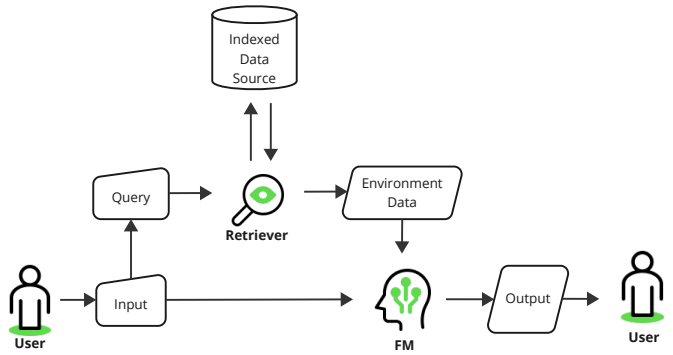
6) Adding a retriever in the generation process not only adds one more AI component to maintain but also additional latency, which can detrimentally impact the time behaviour of the system.

- *Implementation*: The design of the retriever is as important as the design of the FM. For domain-specific use cases, it may be required to train a retriever model on domain-specific data to achieve good suggestions [28]. If latency is a concern, simple retrievers based on term-frequency can be used [34]. When setting up the generation phase using the FM, special care needs to be taken to allow the generation to continue, albeit at lower quality, even if the retriever step fails or the suggestions are of little use. While jointly training the retriever and FM can increase output quality [36], this increases the coupling between both AI components, as the retriever may perform poorly when another FM is used.
- *Related Patterns*: Tool Usage is another approach to allow FMs to access environment data [37]. FMs are trained to make API function calls to access external tools such as search engines and calculators. This approach is more flexible as the FM can call any API available in the environment yet it introduces more security risks as arguments passed to these function calls may result in unauthorized output [32], [33]. A combination of RAG and Tool Usage would be to suggest APIs and API arguments to the FM, but this does not guarantee that the FM will restrict itself to using the suggested arguments. Tool Usage, however, removes the need to have another AI retriever component and maintain indexes of environment data. Another much simpler approach is to simply include the environment data in the prompt, but this is only feasible when this data is static or limited; otherwise the FM prompt can become too large. Fine-tuning the FM can help it memorize environment data but the FM would need to be fine-tuned every time the environment data changes significantly [35].

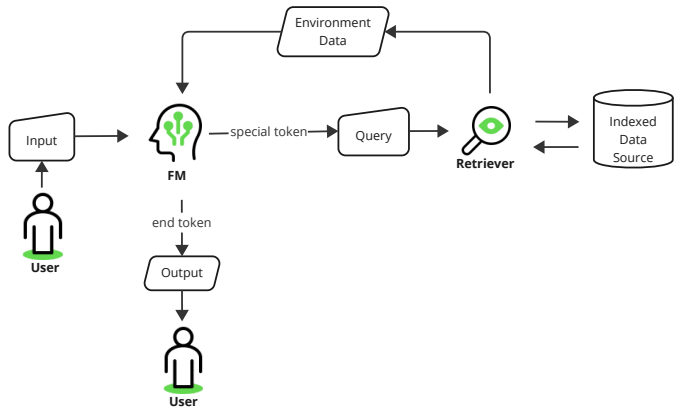
IV. Workflow Generation: A Case Study

Having discussed Task Decomposition and RAG at length, we report how our team built a real-world GenAI application called *Workflow Generation*. We structure our reporting as a descriptive case study [25], where the objective is to show how these design patterns result in a well-engineered AI system.

The context of this case study is a Platform-as-a-Service enterprise company that automates the creation and execution of workflows in domains such as IT and HR. The core team responsible for the GenAI application is made up of a dozen individuals, including software engineers, data labelers, quality engineers, and AI applied researchers. The application



(a) Vanilla RAG where one retrieval takes place per generation.



(b) FM decides when to request for environment data based on outputting special tokens, until an end token is reached; also known as adaptive RAG [21].

Fig. 2: Sample structures of Retrieval-Augmented Generation.

is shipped as an API that is served in the enterprise cloud platform. While another team is responsible for building the user interface, we restrict the case study to only the core team.

There are two research questions that we attempt to answer:

RQ1. *How does Task Decomposition and RAG impact data labeling, model training, model evaluation, and model deployment?*

RQ2. *What are the trade-offs involved when using these two techniques?*

L.E. Lwakatare et al. [6] grouped the software engineering challenges common to commercial AI-based systems in four categories: Assemble dataset, create model, train and evaluate model, and deploy model. Hence, RQ1 is meant to address how these techniques help deal with these challenges in the GenAI era, in terms of software quality attributes.

RQ2 aims to elicit discussion on limitations of these techniques as well as on alternative approaches that were not chosen to build the application. It is easier to understand trade-offs when real-world constraints and requirements are considered.

The sources of information for this case study are mainly of first degree, as the authors of this paper were key contributors in the entire development cycle. Third degree sources,

such as datasets, design documents and experiment journals, complement the self-reporting.

Before answering both research questions, we will define the ML task.

A. ML Task Definition

A workflow automates a repeatable process by listing the steps that need to be executed in a given order. Each of these steps translates to code, but end-users avoid writing code by building a workflow; this is referred to as *low-code*.

Each environment, or installation of the enterprise platform, defines a set of steps $S = \{s_1, s_2, ..., s_n\}$. Each step s has a set of inputs $I_s = \{i_1, i_2, ..., i_n\}$ and a set of outputs $O_s = \{o_1, o_2, ..., o_n\}$. Inputs and outputs have a `name` and a `value`, but the `name` of an input as well as the `name` and `value` of outputs are deterministic (i.e., they do not need to be generated by the FM). Only the `value` of inputs change depending on what the workflow does.

A workflow w is an ordered list of steps $S_w = (w_1, w_2, ..., w_n)$ where every element in S_w comes from set S. In w, the inputs of w_n can use the outputs of w_m where $m < n$, that is, the outputs of previous steps can be used in future steps. Importantly, the input values of w_n can also use environment artifacts such as database table names, table columns and row values.

The ML task then consists of generating workflow w from a natural language text such as *When a P1 incident is created, look up the user assigned to the incident and if the user has a manager, send an email reminding them of the incident*.

Figure 3 shows the expected workflow in YAML format. This representation took several iterations of discussion among team members, as we needed readable workflows for labeling and to make the ML task as intuitive as possible without requiring a large number of tokens. There are four steps:

1) The trigger step determines when the workflow should execute. In this example, it executes when a record in the `incident` database table is created but only if the `priority` value in the record is `1`.
2) The second step has `order: 1` (the trigger step can only be in the beginning of the workflow; thus it has no `order` value). This step reads a record from the `sys_user` table given a condition, which uses an output of the trigger step. Step outputs are represented by having their names inside {{ and }}.
3) The step at `order: 2` tests for a condition using an output from step at `order: 1`.
4) The last step sends an email only if the condition of the previous step is true, as its `block` value is 2, denoting that it is a child of the `IF` step, which has `order: 2`

During our first round of prototyping, we determined that this ML task is very difficult, even with GPT-3.5, a state-of-the-art FM at that point in time. The main challenges are:

- Mapping ambiguous text to specific steps: each serves a particular function and has a unique set of possible inputs.

```yaml
trigger:
  type: record_create
  inputs:
    - name: table
      value: incident
    - name: condition
      value: priority=1
steps:
  - definition: look_up_record
    order: 1
    inputs:
    - name: table
      value: sys_user
    - name: conditions
      value: sys_id={{record_create.current.assigned_to}}
  - definition: IF
    order: 2
    inputs:
    - name: condition
      value: '{{1.Record.manager}}!='
  - definition: send_email
    order: 3
    block: 2
    inputs:
    - name: to
      value: '{{1.Record.manager.email}}'
    - name: subject
      value: {{record_create.current.short_description}}
```

Fig. 3: Sample workflow mapping to the user requirement *When a P1 incident is created, look up the user assigned to the incident and if the user has a manager, send an email reminding them of the incident*. Values coming from the system environment are in light green (prefixed by key `value`).

- Being aware of the workflow's building blocks that are specific to the system environment: steps, database tables, columns, values, etc.
- Understanding the steps' interdependencies: step N can use outputs coming from the previous $N - 1$ steps.
- Generating a complete workflow with acceptable latency: they can be long in number of steps (more than 20), and thus require a large number of tokens.

In order to ship a first version of the application, we had to break the task into two sub-tasks:

1) **createFlow**: Generate the outline of the workflow, listing only the step names and their order/block in the list.
2) **populateInputs**: Generate the inputs of a single step given previous steps.

We then realized that we needed RAG to achieve acceptable quality as the rate of hallucination of step names and database table names was more than 20% in our first experiments. The following subsections will detail how these two decisions to use Task Decomposition and RAG resulted in shippable, well-engineered software.

B. Data Labeling

While labeling is expensive, it is necessary for achieving automatic evaluation. When prompting does not yield good results, the last resort is to fine-tune an FM with labeled

training data. Part of the team tried to write effective prompts, an acknowledged challenge in the software engineering community [14]. The conclusion was that in-context learning via a long description and few-shot samples only worked for simple workflows. We thus decided to fine-tune an FM.

We had thousands of internal workflows that we could use for labeling, to train and evaluate the model. As this is a domain-specific use case, we had to write clear labeling guidelines and train data labelers. To ship an initial version of the GenAI application, we needed to have labeled samples fast. However, training labelers to write a text description (requirement) for a complete workflow such as the one shown in Figure 3 would have taken too long due to the large amount of information present in workflows.

How the task is framed affects heavily the labeling effort. This is where Task Decomposition is useful. When the use case permits it, it is best to break down the labeling effort into multiple rounds where one round maps to a releasable version of the application. In our case, we decided to first ship *Workflow Generation* only with the **createFlow** sub-task.

Our chosen decomposition allowed us to create a dataset for the first task fast so that we could start validating whether a fine-tuned FM could at least generate the outline. Figure 4 shows a sample for the **createFlow** sub-task. Compared to generating the entire workflow, there are two advantages. First, it is much simpler to label as the labeler only had to write a requirement for the steps, ignoring the inputs. Second, it is an easier task for the FM as only the list of steps is generated.

```
requirement: Every day, look up incident tasks and update them.
trigger:
  type: daily
steps:
- definition: look_up_records
  order: 1
- definition: FOREACH
  order: 2
- definition: update_record
  order: 3
  block: 2
```

Fig. 4: Labeling **createFlow** samples involved writing the requirement (in red) given the basic workflow outline.

Figure 5 shows the more complex labeling task for **populateInputs**. In this case, the labeler had to write a label for each step, as an `annotation`, which is normally text that is part of the `requirement`. This was more time-consuming because the labeler needed to understand the values in the step inputs, including column names, column values, and operators. The complexity can be seen even in the simple examples in Figures 4 and 5. The `requirement` evolved from *Every day, look up incident tasks and update them* to *Every day, look up incident tasks that do not have assignees and close them*. The labeler had to map `assigned_toISEMPTY` to *do not have any assignees* and `state=3` to *close them*.

While at the end labelers had to master the intricacies of workflows, Task Decomposition helped phase this learning

```
requirement: Every day, look up incident tasks that do not have
assignees and close them.
trigger:
  annotation: every day
  type: daily
  inputs:
  - name: time
    value: '1970-01-02 00:00:00'
steps:
- annotation: look up incident tasks that do not have assignees
  definition: look_up_records
  order: 1
  inputs:
  - name: table
    value: incident_task
  - name: conditions
    value: assigned_toISEMPTY
- annotation: for each record
  definition: FOREACH
  order: 2
  inputs:
  - name: items
    value: '{{1.Records}}'
- annotation: close them
  definition: update_record
  order: 3
  block: 2
  inputs:
  - name: record
    value: '{{2.item}}'
  - name: table_name
    value: incident_task
  - name: values
    value: state=3
```

Fig. 5: Labeling **populateInputs** samples involved writing one `annotation` (in red) per step where the annotation contains enough detail to generate the `inputs` values (in light green).

into two parts, thereby allowing us to go to market fast and incrementally improve the shipped solution. A trade-off in labeling is that dataset management becomes more complex, as now there are several datasets to verify and maintain.

Task Decomposition also contributed to the modifiability of the system. To continue enhancing the application iteratively, we first asked for labels for the most common steps, as these are simpler to understand. Supporting more steps became simply a matter of labeling less common steps, once the labelers became more experienced.

We did not need any additional labeling for RAG. Data to train the retriever model was obtained by leveraging the same data. The **createFlow** dataset was used to create a RAG dataset for retrieving step names given a user requirement, whereas the **populateInputs** dataset was used for retrieving table names, column names, and column values given an annotation.

C. Model Training

The greatest impact of Task Decomposition and RAG is on model training, or model fine-tuning in our case, as how models are trained determine how they will interface with the rest of the system (i.e., model inputs and outputs). Using samples such as the one shown in Figure 4, we shipped a first version of the application that could generate only outlines, with low rates of hallucination thanks to a simple

RAG implementation. But the real challenge was to generate the complete workflow.

The first step was to find a good enough retriever. As our use case is domain-specific, available open-source retriever models, such as GTR-T5 [38], did not provide satisfactory results. For scalability and latency reasons, we chose a small retriever of about 100 million parameters that encoded text in tens of milliseconds. Using our labeled data, we fine-tuned this small retriever via a common technique called contrastive learning [39], [40], thereby obtaining good retrieval quality.

The next step was to fine-tune the FM to perform the sub-tasks. Given that we were not prompting an off-the-shelf FM, we had flexibility on how to train the FM to do RAG. To decrease latency, it is desirable to minimize the number of retrievals. This is accomplished with a method known as adaptive retrieval [21], where the FM requests additional data, when needed, to continue the generation (Figure 2b). This can result in higher output quality as the FM is trained to rely on the data it has memorized during training for input that is easy or very frequent. For input that is more complex or infrequent in the data distribution, the FM relies on RAG.

Figure 6 shows an example of how the labeled samples for **createFlow** were transformed to suggest choices to the FM coming from the environment. We used `choices:` as a special token to stop generation and call the retriever using the step annotation, in this case: *add work notes to them*. At inference time, when this token is generated, the retriever module is invoked. The samples for **populateInputs** were similarly transformed to give choices to the FM for table names, column names, and column values. Note that, akin to teacher forcing [41], during training the FM will always see the ground-truth value in the list of choices.

```
requirement: Every 5 hours, search for integration records and
add work notes to them
trigger:
  annotation: every 5 hours
  type: repeat
steps:
- annotation: search for integration records
  definition: look_up_records
  order: 1
- annotation: for each
  definition: FOREACH
  order: 2
- annotation: add work notes to them
  choices:
  - add_worknote
  - update_worknote
  - add_workitem
  - add_note
```

Fig. 6: Example of how RAG was used in the training process. Text *choices:* in blue is a special token that the FM uses to signal that it wants suggestions from the environment. Here, the retriever module offers four choices for the step name.

Another important design choice when using Task Decomposition is how to configure the link between the sub-tasks. A crucial difference between the example in Figure 4 and Figure 6, besides *choices:*, is that the latter has a field called *annotation*, which describes the steps. In a later version of the **createFlow** dataset, the FM is fine-tuned to generate these annotations by extracting them from the user requirement. These annotations then become a crucial part of the **populateInputs** training samples, as they are used to populate the environment artifacts in the step inputs.

To orchestrate the sub-tasks, we followed an approach similar to Figure 1b, where the orchestrator is a simple non-AI component that first calls the **createFlow** API. Then it loops over all steps in the outline, generating the inputs by calling the **populateInputs** API for every step. Via this orchestration, the complete flow is generated.

Using Task Decomposition and RAG resulted in three modules: one to generate the outline, one to populate the step inputs, and one to retrieve suggestions. To simplify the system, we use the same fine-tuned FM for both sub-tasks, but the benefits in modularity would allow us to use a larger FM for the more complex problem of generating the outline and a smaller FM for the other task. This would help the scalability of the system because smaller FMs have lower latency and there are many more API calls to populate inputs.

This modularity also gives the system more modifiability and replaceability. We can improve the AI capability to generate the outline independent of the rest of the system and we can retrain the module that populates inputs, once we have more labeling data, so that it supports the less frequent steps.

Decomposing the tasks also helped the time behaviour of the system, as the user can see the outline before deciding to continue the generation. The interaction of the system is further enhanced by allowing the user to stop generation and modify their requirement if the generated outline is erroneous.

A measure of self-descriptiveness, or interpretability, is achieved with the generated annotations, but especially with RAG, as we could show users the choices that were considered by the FM.

Besides the reusability that RAG provides, since the retriever model was fine-tuned for all environment artifacts, RAG improves the security and safety of the system. By using RAG, we ensured that the FM does not contain any customer-specific knowledge in its model weights. The retriever choices can be filtered according to user permissions (confidentiality) and the FM can only use data that the retriever suggests (integrity). Lastly, users can judge the risk of the generated output by consulting the retriever choices (risk identification).

The disadvantages of Task Decomposition in model training is that it becomes multi-task training, which adds complexity such as deciding the proportion of samples for each task and can make training more unstable. Normally a sufficiently large dataset can alleviate these issues. We also had to fine-tune our own retriever model, which required more effort and time. This extra fine-tuning effort can be justified only by the complexity of the workflow generation task.

Figure 7 shows the three-layer architecture of the system, resulting from decisions made in the model training phase. The UI receives the user requirement and displays the workflow

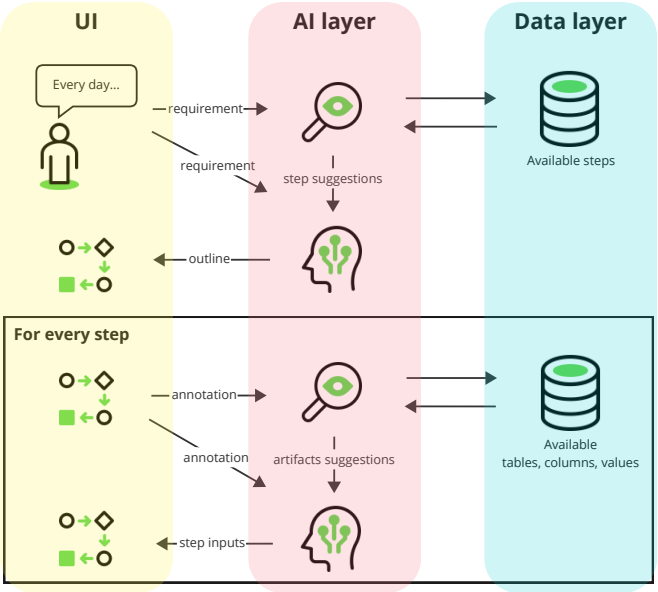


Fig. 7: System architecture with UI, AI, and data layers.

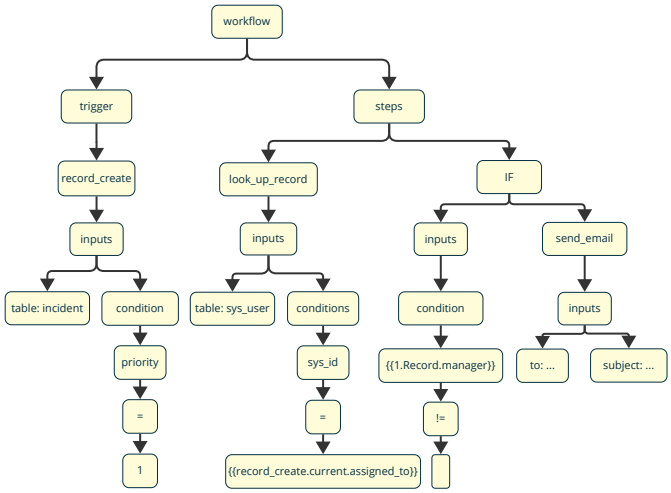


Fig. 8: Tree mapping to the sample workflow in Figure 3.

outline and step inputs as they are populated. The AI layer contains the retriever and the fine-tuned FM. The data layer stores the indexed sources of data, from where the retriever suggests choices to the FM. This layer can be replaced in every installation of the platform to allow the FM to generate output specific to each customer, without compromising security.

D. Model Evaluation

Evaluating the model is as important as data collection. It is generally acknowledged that testing GenAI-based systems is particularly challenging due to the open nature of generation [42]. There is generally more than one generated answer for the same input and retraining FMs can result in different output for the same previously tested input [14]. The simplest yet time-consuming way to evaluate FMs is to use humans knowledgeable of the ML task, while the fastest yet potentially inaccurate approach is to use the strongest available FM as a LLM-as-a-Judge. As using LLMs for evaluation does not always correlate with human evaluations [43] and introduces non-determinism into the evaluation process, we created our own metric that is deterministic and explainable.

The insight behind our metric, called *Flow Similarity*, is that workflows can be represented as trees, similar to how a computer program can be represented as an abstract syntax tree. Once we have the expected and generated workflows as trees, we can use the tree edit distance algorithm [44] to compute a similarity score. We still require humans to give us a expected workflow per user requirement, but we can automate the evaluation.

Figure 8 depicts the tree representation of the workflow shown in Figure 3. Thanks to how we decomposed the tasks, we can evaluate the outline generation and the population of inputs separately, making the system highly testable. We could then decide to enable in the API the population of inputs of only the steps where we had acceptable quality.

Task Decomposition also helps the analysability of the system since we had three groups of evaluation metrics to report to our stakeholders: correctness considering only the outline, correctness considering the entire workflow, and correctness considering only step inputs (e.g., Flow similarity when populating the `look_up_record` step). We rank the steps by how well the FM can populate its inputs and focus on the bottom of the list.

Using RAG also offered testability benefits as we evaluated the retriever separately using recall metrics for every kind of artifact that we needed to retrieve from the environment. We analyzed the errors per kind of artifact to determine that retrieving column names is very difficult as the annotations can be of the form *set it to false*. In this case, the pronoun *it* refers to a previously mentioned column name. Quality can be improved by including more context in the query in addition to the annotation, such as adding the requirement.

Overall, Task Decomposition and RAG contribute to the functional correctness of the system, mainly because they allow testing and improving different parts independently: sub-tasks and the retriever. By systematically testing the different parts, we can more easily discover failure modes. This is similar to the benefits that modularity brings to building traditional software systems as engineers can devote their attention to the underperforming parts.

However, it has been previously noted that ML modularity is not the same as software engineering modularity. Improvements in a sub-task may not necessarily translate to overall system improvements as the rest of the system may not have been tuned according to the improved sub-task output [7]. While using these two patterns enhanced the testability of the GenAI components, we still required significant end-to-end testing.

E. Deployment

The vast size of FMs in number of model parameters creates significant performance and deployment challenges [14]. A flexible architecture can help the deployment process by offering choices. With Task Decomposition, since each sub-task has its own level of ML difficulty and is handled separately, they can be handled by a separate FM to maximize performance.

For instance, the most complex sub-tasks can be assigned to the most powerful FM available and the simpler sub-tasks to smaller FMs, since larger models generally produce better output quality [45]. If the FMs are deployed on local servers, then the smaller FMs can be deployed on smaller, older GPUs such as V100s, reserving the more modern GPUS, A100s and H100s, for larger FMs. Or, if higher quality is needed, the most complex task can be given to a very strong cloud-served FM such as GPT-4o, while the simpler sub-tasks can be served in local servers. To further improve resource utilization, as long as the use case permits it, the sub-tasks can be batched so they can be solved concurrently, as depicted in Figure 1a.

In our case, due to fine-tuning, we were able to obtain good quality with a small FM having only 7 billion parameters. Due to its small size, we had the same FM execute both sub-tasks on a H100 GPU. Unless computing resources are lacking, it is preferable to have the same FM execute all sub-tasks in order to reduce the deployment complexity.

Our RAG encoder is also very small having only 100 million parameters, but it achieves good retrieval quality due to fine-tuning. RAG provides good interoperability as the same encoder (and indexes) can be used for other use cases. As *Workflow Generation* is not the only GenAI capability in the company's platform, the encoder was deployed as an AI component that can be used in other use cases such as Code Generation. Note that this small retriever could run on CPU due to its small size if we did not have GPUs at our disposal.

RAG however makes the deployment more complex as there are two models to deploy: the FM and the retriever encoder. If the Tool Usage pattern is used, only the FM would be deployed. However, the tools or APIs that the FM calls during generation would need to be fixed unless RAG is used to tell the FM what is available in the environment. Current industry practice shows that it is difficult to ship real-world applications with low hallucination rates and with up-to-date knowledge without a RAG implementation.

V. Discussion

The presented case study gives enough detail to answer RQ1 and RQ2. Task Decomposition and RAG resulted in a modular, flexible, secure, and testable system, at the expense of added complexity especially in model training. Breaking down tasks into sub-tasks requires some experimentation and AI knowledge, but it can yield faster prototyping and an iterative path to releasing GenAI applications.

Data labeling still constitutes a significant percentage of the overall human labour in any ML project [14], [46]. Task Decomposition and RAG can add to the labeling effort as each sub-task, including the retrieval task, requires data at least for evaluation. However, dividing labeling tasks into smaller, simpler tasks reduces time-to-market.

Prompting an off-the-shelf FM would have resulted in faster development due to reduced labeling needs and no model training. This is preferable when the ML task is general and not domain-specific. But even when prompting, dividing the use case into sub-tasks can be beneficial. In our case, we had to fine-tune both the FM and the retriever. For more standard GenAI use cases, such as summarization or question-answering, open-source retriever models can achieve good retrieval quality. The crucial advantage with RAG, compared to other approaches that give environment data to an FM, is that it offers more control and thus security over what data can be shown to the FM.

Our model evaluation benefited from Task Decomposition and RAG as we computed evaluation metrics for each task in the GenAI pipeline: the task, the two sub-tasks, and RAG. On the other hand, these two patterns result in more deployment efforts as there are more AI components to deploy.

As our case study is limited by its self-reporting nature, we need to consider the threats to validity, considering that our objective is to show that these two techniques resulted in a well-engineered GenAI system:

- **External validity:** To what extent is it possible to generalize our findings? We framed the case study by first discussing Task Decomposition and RAG in general to motivate their applicability. From our industry experience and participation in AI academic conferences, we judge that the case study will be relevant to the community and speak to current challenges. RAG is a technique that has become very common in the past year and Task Decomposition, while as a term it is not as known as RAG, is simply an application of *divide-and-conquer*.
- **Reliability:** To what extent are the data and the analysis dependent on our own experience? To mitigate this threat, we include as much detail as possible across the main phases of the AI development cycle, including a detailed description of the task and data.

VI. Conclusion

Motivated by our desire to bridge the AI and software engineering communities, we formalize two common techniques, Task Decomposition and Retrieval-Augmented Generation, as GenAI design patterns. They are reusable solutions to common problems and requirements arising from the need to integrate AI components into software systems. We discuss their benefits in terms of software quality attributes and consider their trade-offs and alternatives. Lastly, to illustrate their applicability, we provide a case study of building a real-world GenAI application called *Workflow Generation*.

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REFERENCES

[1] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill, E. Brynjolfsson, S. Buch, D. Card, R. Castellon, N. S. Chatterji, A. S. Chen, K. A. Creel, J. Davis, D. Demszky, C. Donahue, M. Doumbouya, E. Durmus, S. Ermon, J. Etchemendy, K. Ethayarajh, L. Fei-Fei, C. Finn, T. Gale, L. E. Gillespie, K. Goel, N. D. Goodman, S. Grossman, N. Guha, T. Hashimoto, P. Henderson, J. Hewitt, D. E. Ho, J. Hong, K. Hsu, J. Huang, T. F. Icard, S. Jain, D. Jurafsky, P. Kalluri, S. Karamcheti, G. Keeling, F. Khani, O. Khattab, P. W. Koh, M. S. Krass, R. Krishna, R. Kuditipudi, A. Kumar, F. Ladhak, M. Lee, T. Lee, J. Leskovec, I. Levent, X. L. Li, X. Li, T. Ma, A. Malik, C. D. Manning, S. P. Mirchandani, E. Mitchell, Z. Munyikwa, S. Nair, A. Narayan, D. Narayanan, B. Newman, A. Nie, J. C. Niebles, H. Nilforoshan, J. F. Nyarko, G. Ogut, L. Orr, I. Papadimitriou, J. S. Park, C. Piech, E. Portelance, C. Potts, A. Raghunathan, R. Reich, H. Ren, F. Rong, Y. H. Roohani, C. Ruiz, J. Ryan, C. R'e, D. Sadigh, S. Sagawa, K. Santhanam, A. Shih, K. P. Srinivasan, A. Tamkin, R. Taori, A. W. Thomas, F. Tramèr, R. E. Wang, W. Wang, B. Wu, J. Wu, Y. Wu, S. M. Xie, M. Yasunaga, J. You, M. A. Zaharia, M. Zhang, T. Zhang, X. Zhang, Y. Zhang, L. Zheng, K. Zhou, and P. Liang, "On the opportunities and risks of foundation models," *ArXiv*, 2021. [Online]. Available: https://crfm.stanford.edu/assets/report.pdf

[2] H. Zhao, F. Yang, B. Shen, H. Lakkaraju, and M. Du, "Towards uncovering how large language model works: An explainability perspective," 2024. [Online]. Available: https://arxiv.org/abs/2402.10688

[3] Z. Wan, X. Xia, D. Lo, and G. C. Murphy, "How does machine learning change software development practices?" *IEEE Transactions on Software Engineering*, vol. 47, no. 9, pp. 1857–1871, 2021.

[4] L. Heiland, M. Hauser, and J. Bogner, "Design patterns for ai-based systems: A multivocal literature review and pattern repository," in *2023 IEEE/ACM 2nd International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2023, pp. 184–196.

[5] S. J. Warnett and U. Zdun, "Architectural design decisions for the machine learning workflow," *Computer*, vol. 55, no. 3, pp. 40–51, 2022.

[6] L. E. Lwakatare, A. Raj, J. Bosch, H. H. Olsson, and I. Crnkovic, "A taxonomy of software engineering challenges for machine learning systems: An empirical investigation," in *Agile Processes in Software Engineering and Extreme Programming*, P. Kruchten, S. Fraser, and F. Coallier, Eds. Cham: Springer International Publishing, 2019, pp. 227–243.

[7] S. Amershi, A. Begel, C. Bird, R. DeLine, H. Gall, E. Kamar, N. Nagappan, B. Nushi, and T. Zimmermann, "Software engineering for machine learning: A case study," in *2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)*, 2019, pp. 291–300.

[8] J. Jiang, F. Wang, J. Shen, S. Kim, and S. Kim, "A survey on large language models for code generation," 2024. [Online]. Available: https://arxiv.org/abs/2406.00515

[9] H. Washizaki, H. Uchida, F. Khomh, and Y.-G. Guéhéneuc, "Studying software engineering patterns for designing machine learning systems," in *2019 10th International Workshop on Empirical Software Engineering in Practice (IWESEP)*, 2019, pp. 49–495.

[10] S. Martínez-Fernández, J. Bogner, X. Franch, M. Oriol, J. Siebert, A. Trendowicz, A. M. Vollmer, and S. Wagner, "Software engineering for ai-based systems: A survey," *ACM Trans. Softw. Eng. Methodol.*, vol. 31, no. 2, Apr. 2022. [Online]. Available: https://doi.org/10.1145/3487043

[11] X. Franch, S. Martínez-Fernández, C. P. Ayala, and C. Gómez, "Architectural decisions in ai-based systems: An ontological view," in *Quality of Information and Communications Technology*, A. Vallecillo, J. Visser, and R. Pérez-Castillo, Eds. Cham: Springer International Publishing, 2022, pp. 18–27.

[12] A. Serban and J. Visser, " Adapting Software Architectures to Machine Learning Challenges ," in *2022 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. Los Alamitos, CA, USA: IEEE Computer Society, Mar. 2022, pp. 152–163. [Online]. Available: https://doi.ieeecomputersociety.org/10.1109/SANER53432.2022.00029

[13] B. Zhang, T. Liu, P. Liang, C. Wang, M. Shahin, and J. Yu, "Architecture decisions in ai-based systems development: An empirical study," in *2023 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, 2023, pp. 616–626.

[14] A. E. Hassan, D. Lin, G. K. Rajbahadur, K. Gallaba, F. R. Cogo, B. Chen, H. Zhang, K. Thangarajah, G. Oliva, J. J. Lin, W. M. Abdullah, and Z. M. J. Jiang, "Rethinking software engineering in the era of foundation models: A curated catalogue of challenges in the development of trustworthy fmware," in *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering*, ser. FSE 2024. New York, NY, USA: Association for Computing Machinery, 2024, p. 294–305. [Online]. Available: https://doi.org/10.1145/3663529.3663849

[15] R. Cabral, M. Kalinowski, M. T. Baldassarre, H. Villamizar, T. Escovedo, and H. Lopes, "Investigating the impact of solid design principles on machine learning code understanding," in *Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering - Software Engineering for AI*, ser. CAIN '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 7–17. [Online]. Available: https://doi.org/10.1145/3644815.3644957

[16] I. Kumara, R. Arts, D. D. Nucci, W. J. V. D. Heuvel, and D. A. Tamburri, "Requirements and reference architecture for mlops:insights from industry," 2022.

[17] "Iso/iec 25010:2023 systems and software engineering — systems and software quality requirements and evaluation (square) — product quality model," 2023". [Online]. Available: https://www.iso.org/standard/78176.html

[18] C.-Y. Kuan, C.-K. Yang, W.-P. Huang, K.-H. Lu, and H. yi Lee, "Speechcopilot: Leveraging large language models for speech processing via task decomposition, modularization, and program generation," 2024. [Online]. Available: https://arxiv.org/abs/2407.09886

[19] P. Papalampidi, F. Keller, and M. Lapata, "Film trailer generation via task decomposition," 2021. [Online]. Available: https://arxiv.org/abs/2111.08774

[20] J. Kwiatkowski and K. Krawiec, "Learning abstract visual reasoning via task decomposition: A case study in raven progressive matrices," *Int. J. Appl. Math. Comput. Sci.*, vol. 34, no. 2, p. 309–321, Jun. 2024. [Online]. Available: https://doi.org/10.61822/amcs-2024-0022

[21] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," 2024. [Online]. Available: https://arxiv.org/abs/2312.10997

[22] O. Ayala and P. Bechard, "Reducing hallucination in structured outputs via retrieval-augmented generation," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 6: Industry Track)*, Y. Yang, A. Davani, A. Sil, and A. Kumar, Eds. Mexico City, Mexico: Association for Computational Linguistics, Jun. 2024, pp. 228–238. [Online]. Available: https://aclanthology.org/2024.naacl-industry.19

[23] X. Wang, Q. Yang, Y. Qiu, J. Liang, Q. He, Z. Gu, Y. Xiao, and W. Wang, "Knowledgpt: Enhancing large language models with retrieval and storage access on knowledge bases," 2023. [Online]. Available: https://arxiv.org/abs/2308.11761

[24] S. Xu, L. Pang, H. Shen, X. Cheng, and T.-S. Chua, "Search-in-the-chain: Interactively enhancing large language models with search for knowledge-intensive tasks," 2024. [Online]. Available: https://arxiv.org/abs/2304.14732

[25] P. Runeson and M. Höst, "Guidelines for conducting and reporting case study research in software engineering," *Empirical Softw. Engg.*, vol. 14, no. 2, p. 131–164, Apr. 2009. [Online]. Available: https://doi.org/10.1007/s10664-008-9102-8

[26] Q. Lu, L. Zhu, X. Xu, Y. Liu, Z. Xing, and J. Whittle, "A taxonomy of foundation model based systems through the lens of software architecture," in *2024 IEEE/ACM 3rd International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2024, pp. 1–6.

[27] Q. Lu, L. Zhu, X. Xu, Z. Xing, and J. Whittle, "Toward responsible ai in the era of generative ai: A reference architecture for designing foundation model-based systems," *IEEE Softw.*, vol. 41, no. 6, p. 91–100, Jun. 2024. [Online]. Available: https://doi.org/10.1109/MS.2024.3406333

[28] T. Zhang, S. G. Patil, N. Jain, S. Shen, M. Zaharia, I. Stoica, and J. E. Gonzalez, "RAFT: Adapting language model to domain specific RAG," in *First Conference on Language Modeling*, 2024. [Online]. Available: https://openreview.net/forum?id=rzQGHXNReU

[29] Y. Cai, S. Mao, W. Wu, Z. Wang, Y. Liang, T. Ge, C. Wu, W. WangYou, T. Song, Y. Xia, N. Duan, and F. Wei, "Low-code LLM: Graphical user interface over large language models," in

Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: System Demonstrations), K.-W. Chang, A. Lee, and N. Rajani, Eds. Mexico City, Mexico: Association for Computational Linguistics, Jun. 2024, pp. 12–25. [Online]. Available: https://aclanthology.org/2024.naacl-demo.2

[30] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, *Design patterns: elements of reusable object-oriented software*. USA: Addison-Wesley Longman Publishing Co., Inc., 1995.

[31] J. C. M. Tan, P. Saroj, B. Runwal, H. Maheshwari, B. L. Y. Sheng, R. Cottrill, A. Chona, A. Kumar, and M. Motani, "Taskgen: A task-based, memory-infused agentic framework using strictjson," 2024. [Online]. Available: https://arxiv.org/abs/2407.15734

[32] S. Cohen, R. Bitton, and B. Nassi, "A jailbroken genai model can cause substantial harm: Genai-powered applications are vulnerable to promptwares," 2024. [Online]. Available: https://arxiv.org/abs/2408.05061

[33] Z. Wu, H. Gao, J. He, and P. Wang, "The dark side of function calling: Pathways to jailbreaking large language models," 2024. [Online]. Available: https://arxiv.org/abs/2407.17915

[34] S. E. Robertson and S. Walker, "Some simple effective approximations to the 2-poisson model for probabilistic weighted retrieval," in *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR '94. Berlin, Heidelberg: Springer-Verlag, 1994, p. 232–241.

[35] S. Barnett, S. Kurniawan, S. Thudumu, Z. Brannelly, and M. Abdelrazek, "Seven failure points when engineering a retrieval augmented generation system," in *Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering - Software Engineering for AI*, ser. CAIN '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 194–199. [Online]. Available: https://doi.org/10.1145/3644815.3644945

[36] J. Wang, Z. Yang, Z. Yao, and H. Yu, "Jmlr: Joint medical llm and retrieval training for enhancing reasoning and professional question answering capability," 2024. [Online]. Available: https://arxiv.org/abs/2402.17887

[37] T. Schick, J. Dwivedi-Yu, R. Dessí, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: language models can teach themselves to use tools," in *Proceedings of the 37th International Conference on Neural Information Processing Systems*, ser. NIPS '23. Red Hook, NY, USA: Curran Associates Inc., 2024.

[38] J. Ni, C. Qu, J. Lu, Z. Dai, G. Hernandez Abrego, J. Ma, V. Zhao, Y. Luan, K. Hall, M.-W. Chang, and Y. Yang, "Large dual encoders are generalizable retrievers," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, Y. Goldberg, Z. Kozareva, and Y. Zhang, Eds. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, Dec. 2022, pp. 9844–9855. [Online]. Available: https://aclanthology.org/2022.emnlp-main.669

[39] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *2006 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'06)*, vol. 2, 2006, pp. 1735–1742.

[40] T. Gao, X. Yao, and D. Chen, "SimCSE: Simple contrastive learning of sentence embeddings," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 6894–6910. [Online]. Available: https://aclanthology.org/2021.emnlp-main.552

[41] R. J. Williams and D. Zipser, "A learning algorithm for continually running fully recurrent neural networks," *Neural Computation*, vol. 1, no. 2, pp. 270–280, 1989.

[42] B. Abeysinghe and R. Circi, "The challenges of evaluating llm applications: An analysis of automated, human, and llm-based approaches," 2024. [Online]. Available: https://arxiv.org/abs/2406.03339

[43] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, "G-eval: NLG evaluation using gpt-4 with better human alignment," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 2511–2522. [Online]. Available: https://aclanthology.org/2023.emnlp-main.153

[44] K. Zhang and D. Shasha, "Simple fast algorithms for the editing distance between trees and related problems," *SIAM Journal on Computing*, vol. 18, no. 6, pp. 1245–1262, 1989. [Online]. Available: https://doi.org/10.1137/0218082

[45] S. Badshah and H. Sajjad, "Quantifying the capabilities of llms across scale and precision," 2024. [Online]. Available: https://arxiv.org/abs/2405.03146

[46] T. Fredriksson, D. I. Mattos, J. Bosch, and H. H. Olsson, "Data labeling: An empirical investigation into industrial challenges and mitigation strategies," in *Product-Focused Software Process Improvement*, M. Morisio, M. Torchiano, and A. Jedlitschka, Eds. Cham: Springer International Publishing, 2020, pp. 202–216.